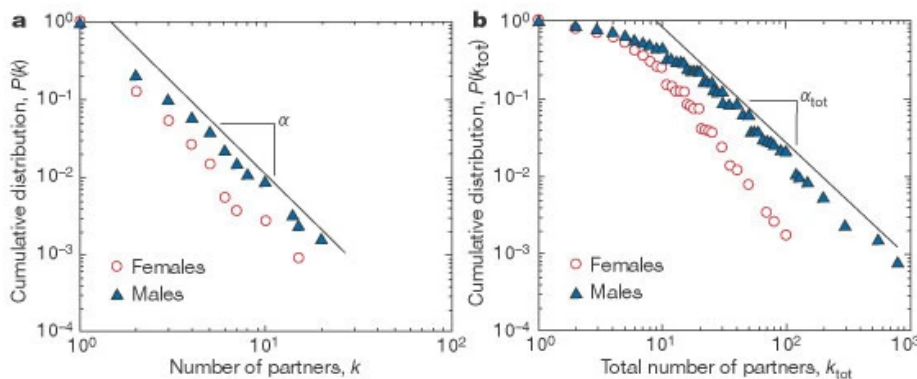


L4: What Does The Power-law Property Mean in Practice?



The focus of this lecture so far has been on the statistical properties of networks with power-law degree distribution. What does this property mean in practice, however? And how does it affect network phenomena that all of us care about, such as the spread of epidemics?

To answer the first question, let us consider the case of networks of sexual partners. There are several diseases that spread through sexual intercourse, including HIV-AIDS, syphilis or gonorrhea. The degree distribution in such networks relates to the number of sexual partners of each individual (*node in the graph*). The plots on this page are based on a 1996 survey of sexual behavior conducted in Sweden. The number of respondents was 2,810 and the age range was from 18 to 74 years old (*roughly balanced between men and women*).

The plot at the left is the C-CDF for the number of partners of each individual during the last 12 months, shown separately for men and women. Note that the distributions drop roughly linearly in the log-log scale plot, suggesting the presence of a power-law distribution (*at least in the range from 2 to 20*).

The plot at the right is the corresponding C-CDF but this time for the entire lifetime of each individual. As expected, the range of the distribution now extends to a wider range (*up to 100 partners for women and 1000 for men*). Note the low-degree saturation effect we discussed earlier in this lesson, especially for less than 10 partners. The exponent of the C-CDF distributions is $\alpha_{tot} = 2.1 \pm 0.3$ for women (*in the range $k_{tot} > 20$*), and $\alpha_{tot} = 1.6 \pm 0.3$ for men (*in the range $20 < k_{tot} < 400$*). Estimates for females and males agree within statistical uncertainty. Note that these exponents refer to the C-CDF – so the corresponding exponents for the degree distributions would be, on average, 3.1 for women and 2.6 for men.

These exponent values suggest that, at least for men, the corresponding network of sexual contacts would have a power-law distribution with very high variability (*theoretically, “infinite variance”*). The distribution also shows the presence of hubs: individuals with hundreds of partners during their lifetime. The wide variability in this distribution justifies targeted intervention approaches that aim to identify the “**hub individuals**” and provide them with additional information, resources (*such as condoms or treatment*), and when available, vaccination.

Image source: <https://www.nature.com/articles/35082140/figures/2>  <https://www.nature.com/articles/35082140/figures/2>